

Circular Economy Ontology Network (CEON) - Actor Module

Metadata

IRI	http://w3id.org/CEON/ontology/actor/
Title	Circular Economy Ontology Network (CEON) - Actor Module
Creator	Huanyu Li
Contributor	Eva Blomqvist Mikael Lindecrantz Robin Keskisärkkä
Date Created	2024-11-13
License	https://creativecommons.org/licenses/by/4.0/
Version Iri	http://w3id.org/CEON/ontology/actor/0.3/
Version Info	0.3
Preferred Namespace Prefix	actor
Preferred Namespace Uri	http://w3id.org/CEON/ontology/actor/
Description	The Actor module of CEON (Circular Economy Ontology Network).

Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESiDE D3.1: C0-1, C0-2, C3-4, T3-4

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESiDE D3.2: C0-1, C0-2, C3-4, T3-4

Classes

Actor Cvn Role ^c	
IRI	http://w3id.org/CEON/ontology/actor/ActorCVNRole
Sub Class Of	ActorCollaborationRole^c
Named Individuals	collectorⁿⁱ dismantlerⁿⁱ manufacturerⁿⁱ recyclerⁿⁱ resellerⁿⁱ sellerⁿⁱ supplierⁿⁱ userⁿⁱ

Actor Collaboration Role^C

IRI	http://w3id.org/CEON/ontology/actor/ActorCollaborationRole
<u>Sub Class Of</u>	actorODP:Role^C
<u>Super Class Of</u>	ActorCVNRole^C ActorProcessRole^C

Actor Process Role^C

IRI	http://w3id.org/CEON/ontology/actor/ActorProcessRole
<u>Sub Class Of</u>	ActorCollaborationRole^C

Actor Resource Role^C

IRI	http://w3id.org/CEON/ontology/actor/ActorResourceRole
<u>Sub Class Of</u>	actorODP:Role^C
<u>Named Individuals</u>	buyerⁿⁱ consumerⁿⁱ issuerⁿⁱ ownerⁿⁱ producerⁿⁱ providerⁿⁱ resellerⁿⁱ sellerⁿⁱ supplierⁿⁱ updaterⁿⁱ userⁿⁱ viewerⁿⁱ

Buying Resource^C

IRI	http://w3id.org/CEON/ontology/actor/BuyingResource
<u>Sub Class Of</u>	actorODP:ResourceRelation^C
<u>Equivalentclass</u>	actorODP:ResourceRelation^C <i>and</i> actorODP:participantRole^{op} value buyer^C

Cvn Participation^C

IRI	http://w3id.org/CEON/ontology/actor/CVNParticipation
<u>Sub Class Of</u>	actorODP:CollaborationParticipation^C actorODP:CollaborationParticipation^C <i>and</i> actorODP:participantRole^{op} some ActorCVNRole^C <i>and</i> actorODP:participationIn^{op} some http://w3id.org/CEON/ontology/cvn/CVN^C <i>and</i> actorODP:participatingActor^{op} some actorODP:Actor^C

Consuming Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/ConsumingResource
Sub Class Of	actorODP:ResourceRelation^C
Equivalentclass	actorODP:participantRole^{OP} value consumer^C and actorODP:ResourceRelation^C

Issuing Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/IssuingResource
Sub Class Of	actorODP:ResourceRelation^C
Equivalentclass	actorODP:ResourceRelation^C and actorODP:participantRole^{OP} value issuer^C

Manufacturing Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/ManufacturingResource
Sub Class Of	actorODP:ResourceRelation^C
Equivalentclass	actorODP:participantRole^{OP} value manufacturer^C and actorODP:ResourceRelation^C

Organisation^C

IRI	http://w3id.org/CE0N/ontology/actor/Organisation
Is Defined By	ISO 59004:2024 - 3.4.1 organization
Description	Person or group of people that has its own functions with responsibilities, authorities, and relationships to achieve its objectives. The concept of organization includes, but is not limited to sole-trader, company, corporation, firm, enterprise, authority, partnership, charity or institution, or part or combination thereof, whether incorporated or not, public or private (e.g. foundation, union, association, agency, municipality, region, country, intergovernmental agencies). A group of organizations can also be considered as an organization that has, alone or collectively, their own objectives.
Sub Class Of	actorODP:Actor^C

Owning Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/OwningResource
Sub Class Of	actorODP:ResourceRelation^C
Equivalentclass	actorODP:ResourceRelation^C and actorODP:participantRole^{OP} value owner^C

Person^C

IRI <http://w3id.org/CE0N/ontology/actor/Person>

Sub Class Of [actorODP:Actor^C](#)

Process Participation^C

IRI <http://w3id.org/CE0N/ontology/actor/ProcessParticipation>

Sub Class Of [actorODP:CollaborationParticipation^C](#)
[actorODP:participatingActor^{op}](#) some [actorODP:Actor^C](#) *and*
[actorODP:participantRole^{op}](#) some [ActorProcessRole^C](#) *and*
[actorODP:CollaborationParticipation^C](#) *and* [actorODP:participationIn^{op}](#) some
[http://w3id.org/CE0N/ontology/processODP/Process^C](http://w3id.org/CE0N/ontology/processODP/Process)

Producing Resource^C

IRI <http://w3id.org/CE0N/ontology/actor/ProducingResource>

Sub Class Of [actorODP:ResourceRelation^C](#)

Equivalentclass [actorODP:ResourceRelation^C](#) *and* [actorODP:participantRole^{op}](#) value [producer^C](#)

Providing Resource^C

IRI <http://w3id.org/CE0N/ontology/actor/ProvidingResource>

Sub Class Of [actorODP:ResourceRelation^C](#)

Equivalentclass [actorODP:participantRole^{op}](#) value [provider^C](#) *and* [actorODP:ResourceRelation^C](#)

Reselling Resource^C

IRI <http://w3id.org/CE0N/ontology/actor/ResellingResource>

Sub Class Of [actorODP:ResourceRelation^C](#)

Equivalentclass [actorODP:ResourceRelation^C](#) *and* [actorODP:participantRole^{op}](#) value [reseller^C](#)

Selling Resource^C

IRI <http://w3id.org/CE0N/ontology/actor/SellingResource>

Sub Class Of [actorODP:ResourceRelation^C](#)

Equivalentclass [actorODP:ResourceRelation^C](#) *and* [actorODP:participantRole^{op}](#) value [seller^C](#)

Interested partyStakeholder^C

IRI	http://w3id.org/CE0N/ontology/actor/Stakeholder
<u>Is Defined By</u>	ISO 59004:2024 - 3.4.2 interested party, stakeholder
<u>Description</u>	Person or organization that can affect, be affected by, or perceive itself to be affected by a decision or activity. To “perceive itself to be affected” means the perception has been made known to the organization.
<u>Sub Class Of</u>	Person^C <i>or</i> Organisation^C

Supplying Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/SupplyingResource
<u>Sub Class Of</u>	actorODP:ResourceRelation^C
<u>Equivalentclass</u>	actorODP:ResourceRelation^C <i>and</i> actorODP:participantRole^{OP} value supplier^C

Updating Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/UpdatingResource
<u>Sub Class Of</u>	actorODP:ResourceRelation^C
<u>Equivalentclass</u>	actorODP:participantRole^{OP} value updater^C <i>and</i> actorODP:ResourceRelation^C

Using Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/UsingResource
<u>Sub Class Of</u>	actorODP:ResourceRelation^C
<u>Equivalentclass</u>	actorODP:participantRole^{OP} value user^C <i>and</i> actorODP:ResourceRelation^C

Viewing Resource^C

IRI	http://w3id.org/CE0N/ontology/actor/ViewingResource
<u>Sub Class Of</u>	actorODP:ResourceRelation^C
<u>Equivalentclass</u>	actorODP:ResourceRelation^C <i>and</i> actorODP:participantRole^{OP} value viewer^C

Actor^C

IRI	http://w3id.org/CE0N/ontology/actorODP/Actor
<u>Super Class Of</u>	Organisation^C Person^C

Capability^C

IRI <http://w3id.org/CEON/ontology/actorODP/Capability>

In Domain Of [capabilityExtent^{op}](#)
[neededResourceRelation^{op}](#)

Collaboration Participation^C

IRI <http://w3id.org/CEON/ontology/actorODP/CollaborationParticipation>

Super Class Of [CVNParticipation^C](#)
[ProcessParticipation^C](#)

Participation^C

IRI <http://w3id.org/CEON/ontology/actorODP/Participation>

Resource Relation^C

IRI <http://w3id.org/CEON/ontology/actorODP/ResourceRelation>

Sub Class Of [actorODP:Participation^C](#) and [actorODP:participantRole^{op}](#) some [ActorResourceRole^C](#) and [actorODP:participatingActor^{op}](#) some [actorODP:Actor^C](#) and [actorODP:participatingResource^{op}](#) some [http://w3id.org/CEON/ontology/resourceODP/Resource^C](http://w3id.org/CEON/ontology/resourceODP/Resource)

Super Class Of [BuyingResource^C](#)
[ConsumingResource^C](#)
[IssuingResource^C](#)
[ManufacturingResource^C](#)
[OwningResource^C](#)
[ProducingResource^C](#)
[ProvidingResource^C](#)
[ResellingResource^C](#)
[SellingResource^C](#)
[SupplyingResource^C](#)
[UpdatingResource^C](#)
[UsingResource^C](#)
[ViewingResource^C](#)

Role^C

IRI <http://w3id.org/CEON/ontology/actorODP/Role>

Super Class Of [ActorCollaborationRole^C](#)
[ActorResourceRole^C](#)

Cvn^C

IRI <http://w3id.org/CEON/ontology/cvn/CVN>

Sub Class Of <http://w3id.org/CEON/ontology/cvn/Collaboration>^C

Collaboration^C

IRI <http://w3id.org/CEON/ontology/cvn/Collaboration>

Super Class Of <http://w3id.org/CEON/ontology/cvn/CVN>^C

Process^C

IRI <http://w3id.org/CEON/ontology/processODP/Process>

Resource^C

IRI <http://w3id.org/CEON/ontology/resourceODP/Resource>

Object Properties

capability extent^{op}

IRI <http://w3id.org/CEON/ontology/actor/capabilityExtent>

Sub Property Of [actorODP:capabilityProperty](#)^{op}

Domain [actorODP:Capability](#)^C

needed resource relation^{op}

IRI <http://w3id.org/CEON/ontology/actor/neededResourceRelation>

Sub Property Of [actorODP:capabilityProperty](#)^{op}

Domain [actorODP:Capability](#)^C

capability property^{op}

IRI <http://w3id.org/CEON/ontology/actorODP/capabilityProperty>

Super Property Of

- [capabilityExtent](#)^{op}
- [neededResourceRelation](#)^{op}

participant role^{op}

IRI <http://w3id.org/CEON/ontology/actorODP/participantRole>

participating actor^{op}

IRI <http://w3id.org/CE0N/ontology/actor0DP/participatingActor>

participating resource^{op}

IRI <http://w3id.org/CE0N/ontology/actor0DP/participatingResource>

participation in^{op}

IRI <http://w3id.org/CE0N/ontology/actor0DP/participationIn>

Annotation Properties

contributor^{ap}

IRI <http://purl.org/dc/terms/contributor>

created^{ap}

IRI <http://purl.org/dc/terms/created>

creator^{ap}

IRI <http://purl.org/dc/terms/creator>

description^{ap}

IRI <http://purl.org/dc/terms/description>

license^{ap}

IRI <http://purl.org/dc/terms/license>

title^{ap}

IRI <http://purl.org/dc/terms/title>

preferred namespace prefix^{ap}

IRI <http://purl.org/vocab/vann/preferredNamespacePrefix>

preferred namespace uri^{ap}

IRI <http://purl.org/vocab/vann/preferredNamespaceUri>

covers requirements^{ap}

IRI <http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements>

Namespaces

:

<http://w3id.org/CEON/ontology/actor/>

actorODP

<http://w3id.org/CEON/ontology/actorODP/>

dcterms

<http://purl.org/dc/terms/>

odp

<http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#>

owl

<http://www.w3.org/2002/07/owl#>

prov

<http://www.w3.org/ns/prov#>

rdf

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

rdfs

<http://www.w3.org/2000/01/rdf-schema#>

vann

<http://purl.org/vocab/vann/>

Legend

^c	Classes
^{op}	Object Properties
^{ap}	Annotation Properties